



ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

AGRICULTURAL TECHNOLOGY

PAPER 1

MARKS: 100

TIME: 2 hours

Topics: safety, materials and tools, fencing and farm structures, energy, and water supply.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions. Show ALL calculations, including formulae and units.

QUESTION 1

- 1.1** Choose the correct answer. Write only the letter (A–D).

- 1.1.1 Which protective equipment protects the eyes during arc welding?

A Ear plugs	B A welding helmet with a dark filter lens
C A dust mask	D Leather gloves

- 1.1.2 Galvanised steel is coated with ... to prevent rust.

A paint	B zinc
C copper	D tin

- 1.1.3 The tool used to tighten fencing wire is a ...

A wire strainer.	B hacksaw.
C spirit level.	D chisel.

- 1.1.4 Which energy source is renewable?

A Coal	B Diesel
C Solar	D Natural gas

- 1.1.5 The unit of electrical power is the ...

A volt.	B ampere.
C watt.	D ohm.

(5 × 2)
(10)

- 1.2** Give ONE term for each description.

- 1.2.1 A mixture of two or more metals, such as brass or bronze

- 1.2.2 The strong post at the end of a fence line that holds the strain of the wires

- 1.2.3 A deep, narrow hole drilled to reach underground water

- 1.2.4 Joining two metal pieces by melting them together

- 1.2.5 A fence that gives animals a short, safe electric shock to keep them in a camp

(5 × 2)
(10)

[20]

QUESTION 2: SAFETY, MATERIALS AND TOOLS

- 2.1** Give FIVE general safety rules for working in a farm workshop. (5)

- 2.2** Distinguish between ferrous and non-ferrous metals. Give ONE example of each. (4)

- 2.3** Explain why steel rusts, and give TWO ways to prevent rust. (4)

- 2.4** Name TWO hand tools used for cutting metal and ONE tool used for measuring. (3)

- 2.5 Give FOUR guidelines for looking after (maintaining) hand tools. (4)
- [20]

QUESTION 3: FENCING AND FARM STRUCTURES

- 3.1 Give THREE reasons why fences are important on a farm. (3)
- 3.2 A farmer wants to fence a rectangular camp of 400 m by 250 m with 5 strands of wire.
- 3.2.1 Calculate the perimeter of the camp. (2)
- 3.2.2 Calculate the total length of wire needed. (2)
- 3.2.3 Droppers are placed every 5 m. Calculate how many droppers are needed. (2)
- 3.2.4 Wire costs R1 250 per 500 m roll. Calculate the cost of the wire. (3)
- 3.3 Describe the steps in erecting a straining post (end post). (4)
- 3.4 Name FOUR materials used to build farm structures such as sheds and kraals. (4)
- [20]

QUESTION 4: ENERGY ON THE FARM

- 4.1 Name FOUR sources of energy that a farm can use, and say whether each is renewable or non-renewable. (8)
- 4.2 An electric water pump has a power rating of 750 W and runs for 6 hours per day. Electricity costs R2,50 per kWh.
- 4.2.1 Calculate the energy used per day in kWh. (3)
- 4.2.2 Calculate the cost of running the pump for 30 days. (3)
- 4.3 Give THREE advantages of using solar panels on a farm. (3)
- 4.4 Give ONE safety rule when working with electricity on a farm. (3)
- [20]

QUESTION 5: WATER SUPPLY

- 5.1 Name FOUR sources of water for a farm. (4)
- 5.2 Name TWO types of water pumps used on farms. (2)
- 5.3 A cylindrical water tank has a diameter of 2 m and a height of 1,5 m. Calculate its volume in litres. Use $\pi = 3,14$ and $1 \text{ m}^3 = 1\,000 \text{ l}$. (5)
- 5.4 Give FOUR ways to save water on a farm. (4)
- 5.5 Explain why water for animals should be tested regularly. (5)
- [20]

TOTAL: 100

END OF QUESTION PAPER